7.4 Difference Between Mitosis and Meiosis

Table 7.3: Difference between mitosis in Plants and Animals

Plants	Animals
Centrioles are absent	Centrioles are present
Asters are not formed	Asters are formed
Cell division involves formation of a cell plate	Cell division involves furrowing and cleavage of cytoplasm
Occurs mainly at meristem	Occurs in tissues throughout the body

Table 7.4: Difference Between Mitosis and Meiosis (Figure 7.8)

Mitosis	Meiosis
One division	Two divisions
Number of chromosomes remains the same	Number of chromosomes is halved
Homologous chromosomes line up separately on the metaphase plate	Homologous chromosomes line up in pairs at the metaphase plate
Homologous chromosome do not pair up	Homologous chromosome pairup to form bivalent
Chiasmata do not form and crossing over never occurs	Chiasmata form and crossingover occurs
Daughter cells are genetically identical	Daughter cells are genetically different from the parent cells
Two daughter cells are formed	Four daughter cells are formed

7.5 Mitogens

The factors which promote cell cycle proliferation is called **mitogens**. Plant mitogens include gibberellin, ethylene, Indole acetic acid, kinetin. These increase mitotic rate.

Mitotic Poisons (Mitotic Inhibitors)

Certain chemical components act as inhibitors of the mitotic cell division and they are called **mitotic poisons**.

Endomitosis

The replication of chromosomes in the absence of nuclear division and cytoplasmic division resulting in numerous copies within each cell is called **endomitosis**. Chromonema do not separate to form chromosomes, but remain closely associated with each other. Nuclear membrane does not rupture. So no spindle formation. It occurs notably in the salivary glands of *Drosophila* and other flies. Cells in these tissues contain giant chromosomes (polyteny), each consisting of over thousands of intimately associated, or synapsed, chromatids. Example: Polytene chromosome.

Anastral

This is present only in plant cells. No asters or centrioles are formed only spindle fibres are formed during cell division.

Amphiastral

Aster and centrioles are formed at each pole of the spindle during cell division. This is found in animal cells.